CSE350: Digital Electronics & Pulse Techniques

Lecture 11: Op Amp Comparators, Schmitt Trigger

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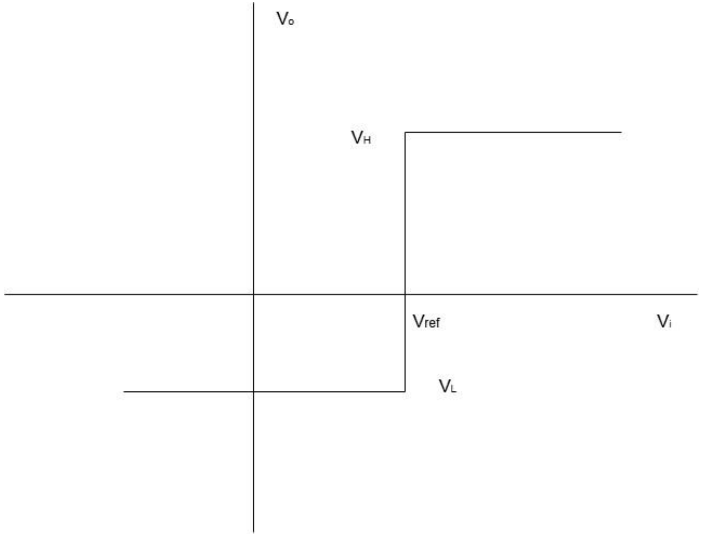
Op Amp Comparator:

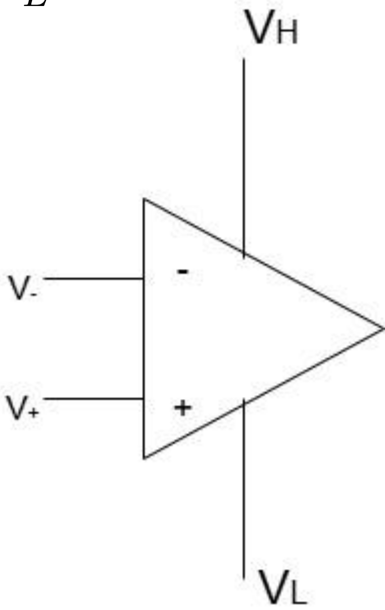
When the non-inverting input V_+ is higher than the inverting input V_-, the output of the op-amp is the higher bias voltage V_H. When it is the opposite scenario, the output is the low bias voltage V_L.

To use the op-amp as a comparator, we usually apply a reference voltage to one of its terminals, and an input voltage to the other. Thus there can be two cases:

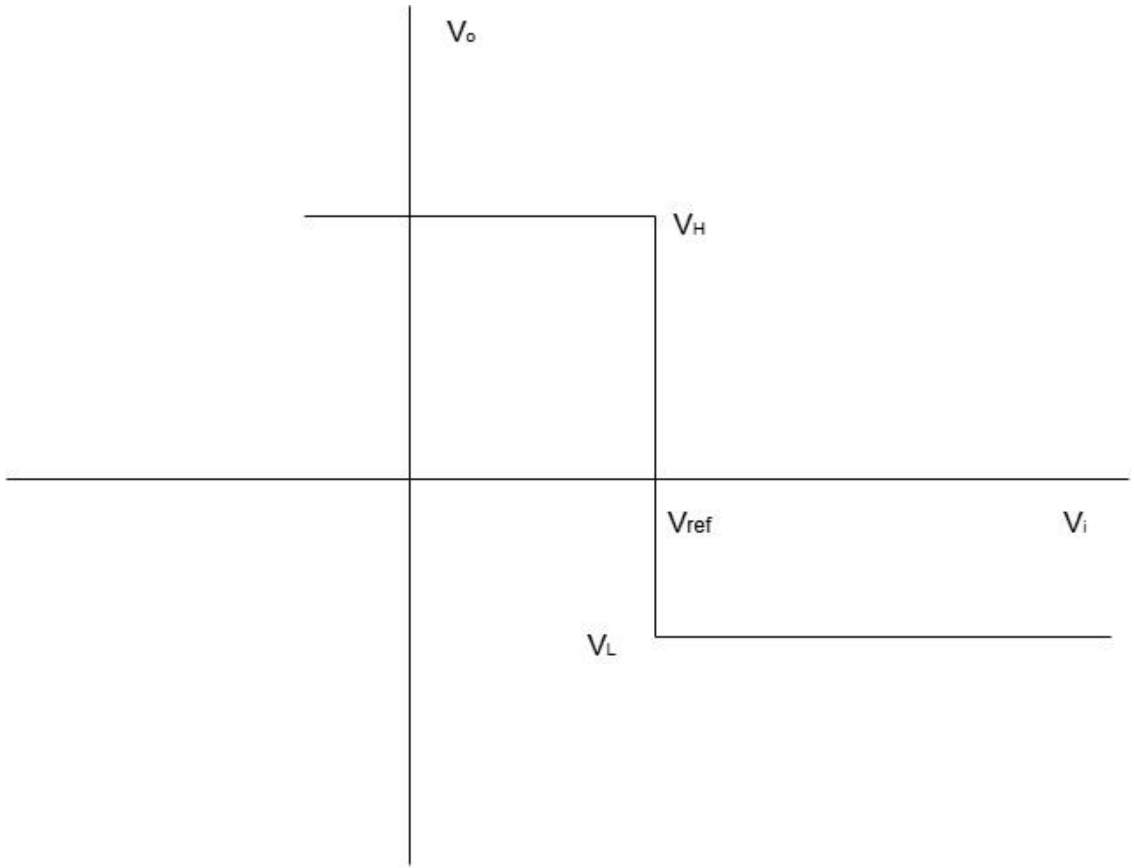
Case 1: Input voltage at non-inverting terminal & reference voltage at inverting terminal.

The output voltage will vary as follows:



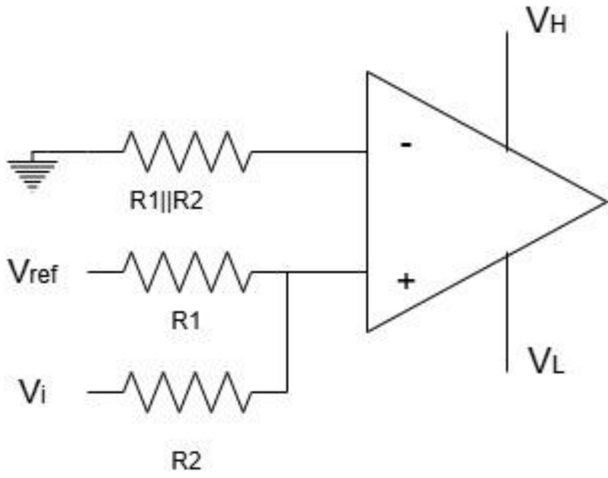


Case 2: Input voltage at inverting terminal & reference voltage at non-inverting terminal.



Let us now consider the following case with resistances at inputs. It is a common practice to have same resistance at both terminals to avoid input bias current errors in op-amps.

Now, since the current at inverting terminal is 0, the voltage will also be 0. Let us now determine the voltage at non-inverting terminal through input & reference voltage.



Applying KCL, $\dfrac{V_{ref} - V_+}{R_1} = \dfrac{V_+ - V_i}{R_2}$

$$V_+ = V_i \frac{R_1}{R_1 + R_2} + V_{ref} \frac{R_2}{R_1 + R_2}$$

$V_+ > V_-$ then the output will be V_H

$$V_i \frac{R_1}{R_1 + R_2} + V_{ref} \frac{R_2}{R_1 + R_2} > 0$$

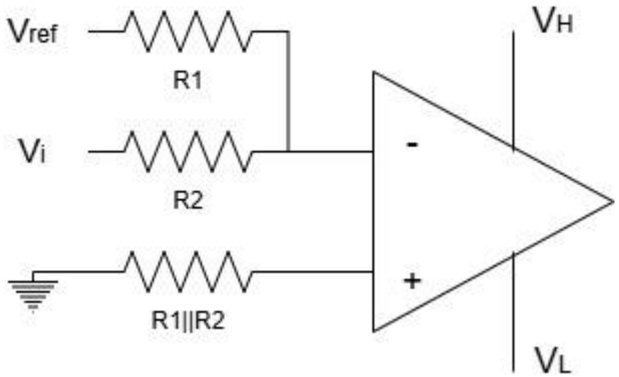
$$V_i > -\frac{R_2}{R_1} V_{ref}$$

Through a similar process, we can show for the case shown,

$V_i < -\frac{R_2}{R_1}V_{ref}$, then the output will be high.

The above two cases can be used to change transition voltage

by tuning R1 & R2 resistors.



Schmitt Trigger Circuit:

The operation of the circuit largely depends on initial conditions. Let us assume, V_i is a negative voltage with a large magnitude. Since no current flows through the inverting terminal,

$$V_i = V_-$$

Applying KCL at non-inverting terminal,

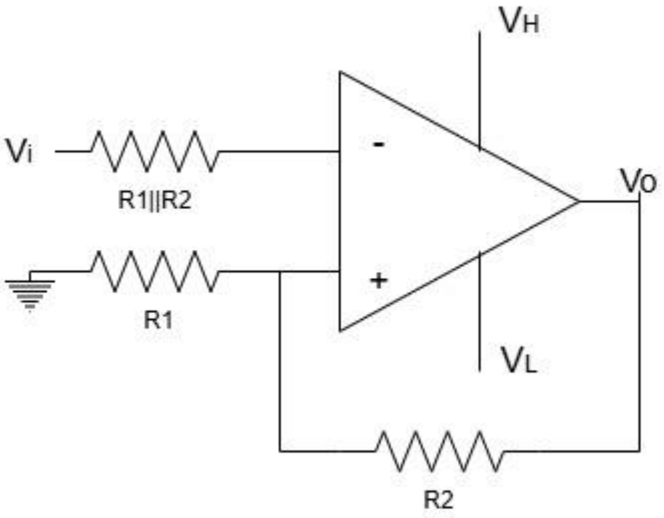
$$\frac{V_+ - 0}{R_1} + \frac{V_+ - V_o}{R_2} = 0$$

$$V_+ = \frac{R_1}{R_1 + R_2} V_o$$

We may assume that initially, $V_o = V_H$ since, the input was a negative voltage. Then the output will remain this way until,

$$V_- < V_+ = V_H \frac{R_1}{R_1 + R_2}$$

If this value is crossed, output will become V_L



Thus this voltage is called high threshold voltage, V_{TH}

$$V_{TH} = V_H \frac{R_1}{R_1 + R_2}$$

Now if we consider the case when the input was initially a large positive voltage, we can assume that the output would be V_L, since the inverting terminal voltage would be higher. The output will remain this way, until,

$$V_- > V_+ = V_L \frac{R_1}{R_1 + R_2}$$

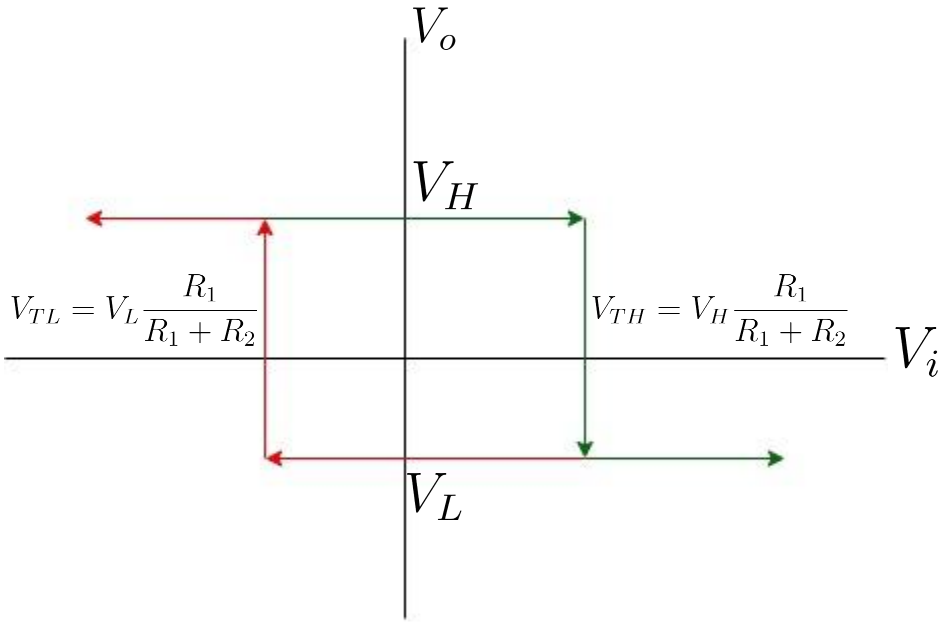
This is the low threshold voltage, V_{TL}

$$V_{TL} = V_L \frac{R_1}{R_1 + R_2}$$

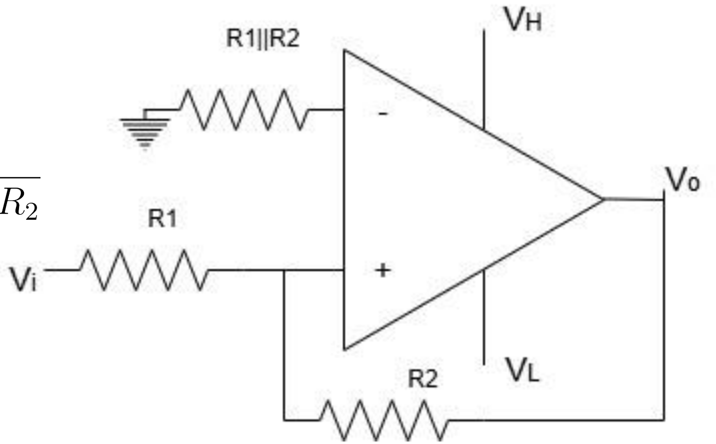
We can combine these two cases and get the transfer characteristics of Schmitt Trigger:

The difference between the two threshold voltages is called ‘Hysteresis Width’

This kind of Schmitt trigger is known as ‘Inverting Schmitt trigger’ because, when input is very negative/positive, output is positive/negative respectively.



Non-inverting Schmitt Trigger:



Applying KCL at non-inverting terminal, $V_+ = V_i \frac{R_2}{R_1 + R_2} + V_o \frac{R_1}{R_1 + R_2}$

$$\frac{V_+ - V_i}{R_1} + \frac{V_+ - V_o}{R_2} = 0$$

If the input is assumed to be a large negative voltage initially, then output will be V_L until input equals higher threshold voltage V_{TH}. [Since, $V_- = 0$]

At transition point, $V_+ = V_- = 0$

And, $V_i = V_{TH}$

So,

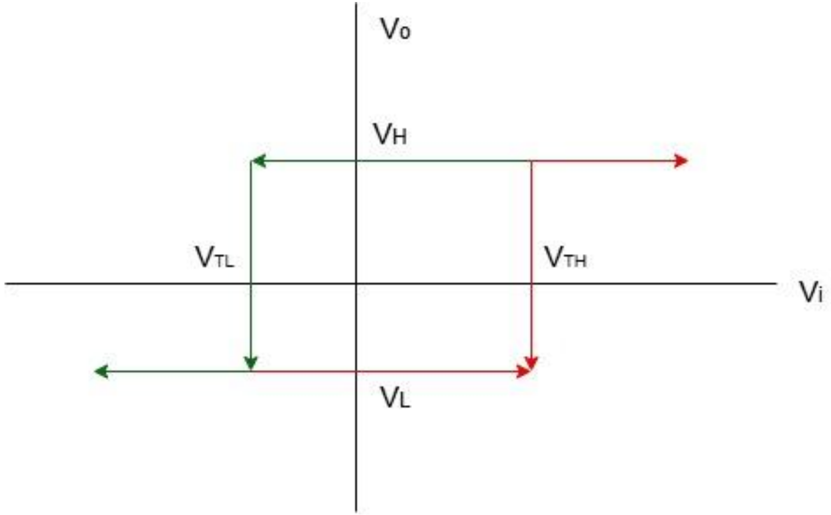
$$0 = V_{TH} \frac{R_2}{R_1 + R_2} + V_L \frac{R_1}{R_1 + R_2}$$

Thus, we get,

$$V_{TH} = -\frac{R_1}{R_2} V_L$$

Similarly,

$$V_{TL} = -\frac{R_1}{R_2} V_H$$



Schmitt Trigger with Applied Voltage:

Inverting Schmitt Trigger:

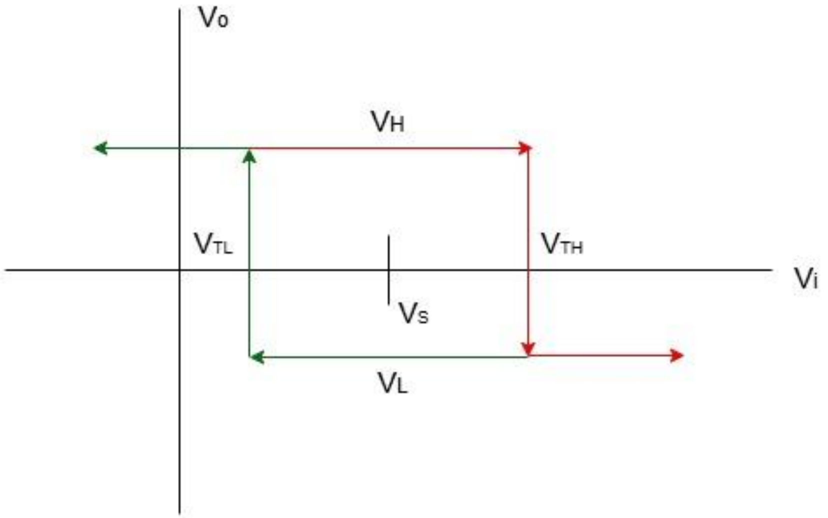
An analysis similar to the previous circuit will give,

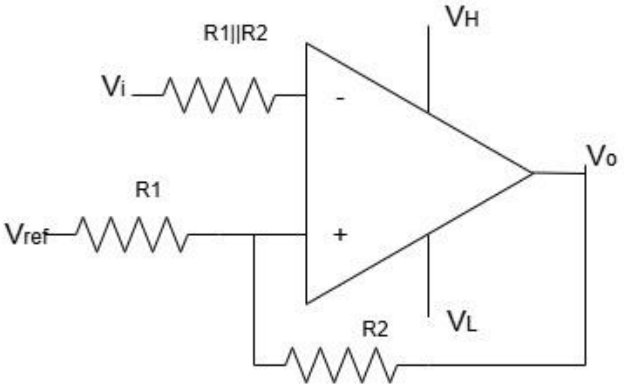
$$V_{TH} = V_{ref}\frac{R_2}{R_1 + R_2} + V_H\frac{R_1}{R_1 + R_2}$$

$$V_{TL} = V_{ref}\frac{R_2}{R_1 + R_2} + V_L\frac{R_1}{R_1 + R_2}$$

We see that the amount of shift in the thresholds is,

$$V_s = V_{ref}\frac{R_2}{R_1 + R_2}$$





Non-inverting Schmitt Trigger:

Through a similar analysis,

$$V_{TH} = \frac{R_1 + R_2}{R_2} V_{ref} + (-\frac{R_1}{R_2}) V_L$$

$$V_{TL} = \frac{R_1 + R_2}{R_2} V_{ref} + (-\frac{R_1}{R_2}) V_H$$

Thus, the shift voltage is,

$$V_s = \frac{R_1 + R_2}{R_2} V_{ref}$$